

Homework 4

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- 1) 1. (5pts) At room temperature, which semiconductor has the largest intrinsic carrier concentration? Explain.
(a) Si ($E_g = 1.1$ eV) (b) Ge ($E_g = 0.7$ eV) (c) GaAs ($E_g = 1.4$ eV)

Correct Answer b) Ge ($E_g = 0.7$ eV)

- Bandgap (E_g) & n_i relationship

$$n_i \propto T^{3/2} \exp\left(-\frac{E_g}{2k_B T}\right)$$

The relationship shows that GaAs (1.4 eV) has a very large gap, very few intrinsic carriers.

- 2) 2. (5pts) At room temperature, which semiconductor has the highest electron mobility? Explain.
(a) Si ($N_D = 10^{14} / \text{cm}^3$) (b) Si ($N_D = 10^{17} / \text{cm}^3$) (c) Both (a) and (b) have same mobility.

Answer: a) Si ($N_D = 10^{14} / \text{cm}^3$) low doping levels (light doped). There are fewer ionized impurities, meaning electrons encountered fewer scattering events & can move freely.

- 3) b) Si ($N_D = 10^{17} / \text{cm}^3$) has the highest conductivity for an n-type semiconductor since it is defined by the equation:

$$\sigma = q N_D \mu_n \quad \& \quad \text{Sample b has 1000 times more electrons than Sample a.}$$

- 4) 4. (15pts) At any temperature, the mobility of n-type GaAs is higher than that of n-type Si. Make a back of the envelope estimate of the ratio of the two mobilities, and compare the value to the actual ratio from the graphs in the lecture notes at $T = 300\text{K}$. Use values for the effective masses given here:
<http://www.ioffe.ru/SVA/NSM/Semicond/Si/basic.html> and
<http://www.ioffe.ru/SVA/NSM/Semicond/GaAs/basic.html>.
Note, from the values given, you will first need to determine the conductivity effective masses. These may not be the same as the density of states effective mass. See slide 25 of 'Slides L1' linked in Week 4 Lectures.

$$\frac{\mu_n \text{ GaAs}}{\mu_n \text{ Si}} \approx \frac{m_n^* \text{ Si}}{m_n^* \text{ GaAs}} = \frac{0.26 m_0}{0.026 m_0} \approx 10$$

$$\frac{\mu_n \text{ GaAs}}{\mu_n \text{ Si}} = \frac{8000}{1358} \approx 5.89 \quad @ \quad T = 300\text{K} \quad \begin{matrix} 8000 \text{ cm}^2/\text{Vs} \rightarrow \text{GaAs} \\ 1358 \text{ cm}^2/\text{Vs} \rightarrow \text{Si} \end{matrix}$$

5) Photon Scattering

6) Average Time between collisions

$$T = \frac{m_n^* \mu_n^*}{q} = 0.3 \cdot 10^{-13} \text{ s}$$

$$M = 8000 \text{ cm}^2/\text{Vs} = 0.08 \text{ (m}^2/\text{Vs)} \\ m_0 = 9.11 \times 10^{-31} \text{ Kg}$$

- 7) 7. (10pts) If three different scattering processes are present in GaAs with scattering lifetimes of $\tau_1 = 10$ ps, $\tau_2 = 20$ ps, and $\tau_3 = 40$ ps, estimate the n-type mobility. Use $m_n^* = 0.067 m_0$.

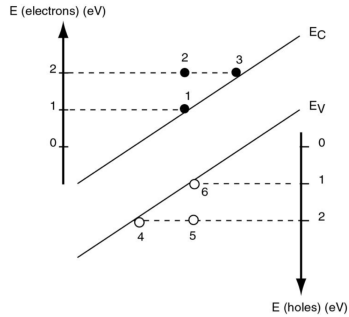
Mattiesen's Rule

$$\frac{1}{\mu_{\text{total}}} = \frac{m_n^*}{q} \left(\frac{1}{\tau_1} + \frac{1}{\tau_2} + \frac{1}{\tau_3} \right) = \frac{m_n^*}{q} \left(\frac{1}{40 \times 10^{-12}} + \frac{1}{20 \times 10^{-12}} + \frac{1}{10 \times 10^{-12}} \right) = 150000 \text{ cm}^2/\text{Vs}$$

8)

8. Given the following energy band diagram, fill in the table specifying the kinetic and potential energies of the numbered particles (2pts each)

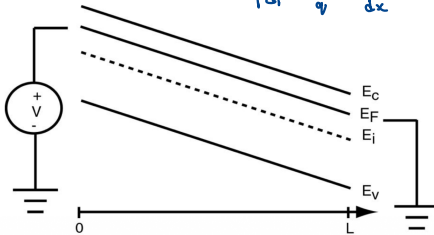
Carrier	Kinetic Energy (eV)	Potential Energy (eV)
electron 1	0	2
electron 2	1	1
electron 3	$2-2=0$	2
hole 4	0	0
hole 5	$2-1=1$	1
hole 6	0	1



9)

9. On the band diagram below, a bias is applied across the semiconductor of length $L=1\text{cm}$. $E_f(x=0) = 2\text{ eV}$. $E_f(x=L) = 0\text{ eV}$. Everywhere, $E_f - E_i = 0.3\text{ eV}$.

$$|E| = \frac{1}{q} = \frac{\Delta E_f}{dx} = \frac{1}{q} = \underline{2\text{ V/cm}}$$



- (a) Is the applied voltage positive or negative with respect to ground? Why?
Negative, since it would have a downward slope

- (b) What is the magnitude and direction of the electric field? Explain.
left, 2 V/cm it points from high to low potential

- (c) What is the direction of the electron diffusion current?

$$n = n_i e^{(E_f - E_i) / k_B T}$$

$$n A E_f - E_i = \text{constant} \rightarrow \underline{\text{zero}}$$

- (d) What is the direction of the hole diffusion current?

$$\text{Zero, } J_{\text{diff}} \propto \frac{dn}{dx} \rightarrow n \text{ \& } p \text{ are both constant}$$

- (e) What is the direction of the electron drift current?

going left

- (f) What is the direction of the hole drift current?

going left

- (g) What is the direction of the total current?

left

10)

10. Assuming the material in problem (9) is Si at $T=300\text{K}$ with $n_i = 10^{10}/\text{cm}^3$, what is the

(a) (10pts) electron density,

$$n = n_i e^{(E_F - E_i) / k_B T} \quad n = 10^{19} e^{0.3 / 0.03} = \underline{1.03 \times 10^{16} / \text{cm}^3}$$

(b) (10pts) mobility $\left(\frac{\text{cm}^2}{\text{Vs}}\right)$,

$$\mu = 1345 \text{ cm}^2/\text{Vs}$$

(c) (10pts) conductivity $\left(\frac{1}{\Omega \cdot \text{cm}}\right)$, and

$$q \mu_n n = 0.222 \text{ } 1/\Omega \cdot \text{cm}$$

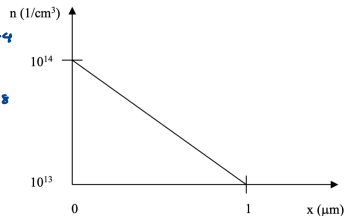
(d) (10pts) magnitude of the current density $\left(\frac{\text{A}}{\text{cm}^2}\right)$?

$$J = G E = 0.222 (2) \Rightarrow \underline{0.444 \text{ A/cm}^2}$$

11)

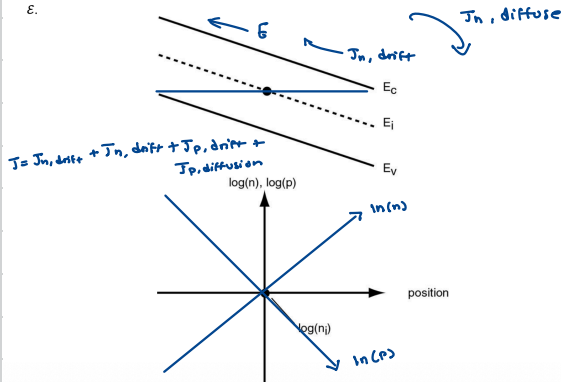
11. (20pts) The electron density in Si at $T=300\text{K}$ in the region between $x=0$ and $x=1 \mu\text{m}$ is as shown in the figure below. The electric field in that region is zero. What is the current, J_n , due to the electrons? Give both the magnitude (A/cm^2) and sign. Use parameters appropriate for a carrier concentration of $n = 10^{14}/\text{cm}^3$ from graph on slide 21 of "Wk4_L1_W24.pdf" linked under Week 4, Slides L1.

$$\begin{aligned} J_n &= q D_n \frac{dn}{dx} & \frac{D_n}{\mu_n} &= \frac{kT}{q} \\ &= \mu_n T_e \frac{dn}{dx} & &= -9 \cdot 10^9 \text{ cm}^2/\text{Vs} \\ &\quad \downarrow 1358 & D_n &= 0.026 \cdot 1358 \\ & & &= 35.31 \\ &\underline{-5.086 \text{ A/cm}^2} \end{aligned}$$



12)

12. Consider the following *equilibrium* band diagram for a portion of a semiconductor sample with a built-in electric field $\mathcal{E}$.



(a) (5pts) Sketch the Fermi level as a function of position through the indicated point across the width of the band diagram above.

(b) (5pts) On the band diagram, sketch the direction of the electric field. Is the field constant or position dependent?

(c) (5pts) Show with an arrow labeled ($J_{N-diffusion}$) the direction of the electron diffusion current. Write down the appropriate equation.

$$J_{N-diffusion} = q D_n \frac{dn}{dx}$$

(d) (5pts) Show with an arrow labeled ($J_{H-diffusion}$) the direction of the hole diffusion current. Write down the appropriate equation.

$$J_{H-diffusion} = -q D_p \frac{dp}{dx}$$

(e) (5pts) Show with an arrow labeled ($J_{H-drift}$) the direction of the hole drift current. Write down the appropriate equation.

$$J_{H-drift} = q p \mu_p \mathcal{E}$$

(f) (5pts) Show with an arrow labeled ($J_{N-drift}$) the direction of the electron drift current. Write down the appropriate equation.

$$J_{N-drift} = q n \mu_n \mathcal{E}$$

(g) (5pts) What is the direction of the total current? Why? $\rightarrow$ zero because E_F is flat

(h) (5pts) On the graph, sketch and label both the electron and hole concentrations as a function of position across the full width of the sample. Note, you are sketching the natural log of the carrier densities versus position. Show the equations that you are plotting.

$$n = n_i e^{(E_F - E_i) / kT}$$

$$\ln(n) \propto (E_F - E_i)$$

$$\ln(p) \propto (E_i - E_F)$$

13)

13. (10 pts) The spec sheet gives the sheet resistance of a diffusion layer as 200Ω . You need to layout a $1 \text{ k}\Omega$ resistor that is $1 \mu\text{m}$ wide on chip. How long must it be?

$$L = \frac{R}{R_s} w = \frac{1000}{200} 1 \mu\text{m} = \underline{\underline{5 \mu\text{m}}}$$